

LDscnR — LD-aware outlier regions

Finding the loci underlying adaptation by using linkage disequilibrium as evidence rather than as a nuisance.

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github.com/genomeinsights/LDscnR

THE PREMISE

Genome scans confuse signal with structure. Small samples, complex demography and relatedness all produce peaks that look like selection. **LD tells them apart.** Selection locally inflates linkage disequilibrium — a causal variant drags a block of correlated markers with it — so a true signal appears as a *cluster* of mutually correlated markers, while a false positive arising from relatedness sits isolated, with low LD to its neighbours.

“LD-supported” and “clustered” are therefore not two separate criteria but the same footprint, seen once per marker and once per region. Everything below follows from that.

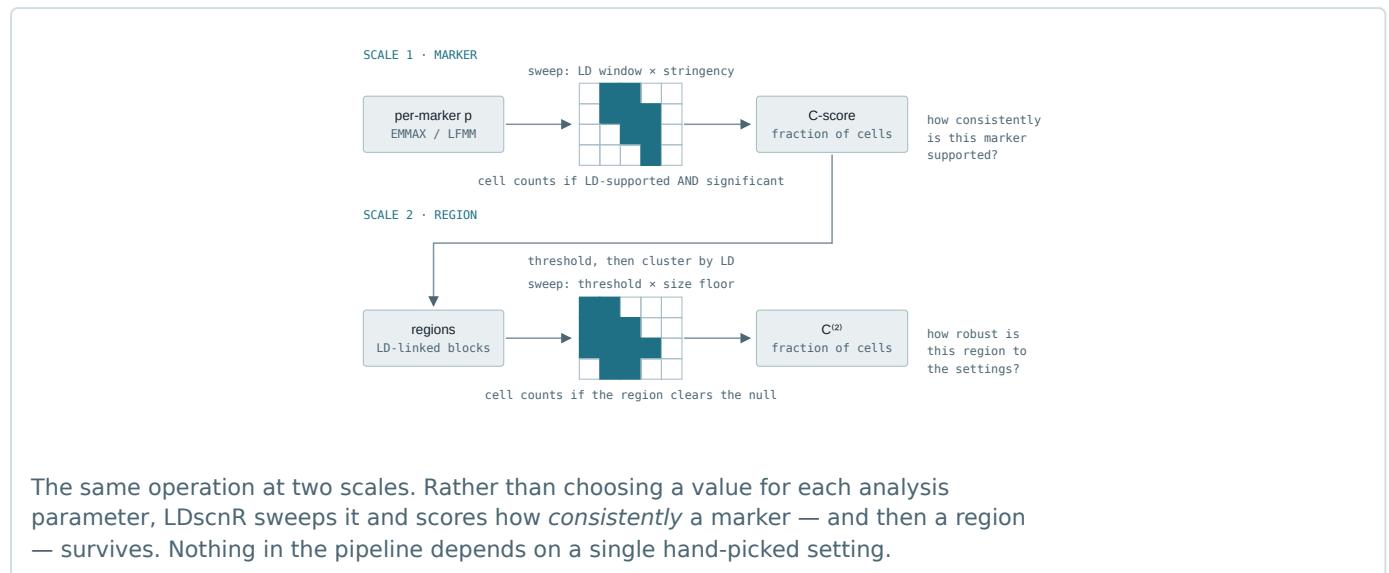
THE PIPELINE

00 Detectability Can this genome be scanned at all? A truth-free check that signal is separable from background, so structure-dominated datasets are flagged rather than over-interpreted.	01 Kinship Build the relatedness matrix from LD-independent markers, so it cannot absorb the very regions under selection. Stringency set by the genomic inflation factor.
02 Association scan EMMAX, LFMM, or any test giving one <i>p</i> -value per marker. The method consumes the vector and is agnostic to its source.	03 Marker C-score Sweep the LD window and the filtering stringency. C = the fraction of settings in which a marker is both LD-supported and significant.
04 Outlier regions Cluster surviving markers by <i>linkage</i> , not by physical distance — so an inversion stays one unit while an adjacent looser cluster separates.	05 Structure-aware null Surrogate phenotypes that keep the population structure but break the link to the environment. Each region gets a location-matched, FDR-controlled <i>p</i> -value.
06 Second-tier C-score Sweep the region threshold and the size floor. C⁽²⁾ = the fraction of settings in which a locus stays significant — a robustness ranking.	07 Report Ranked loci with two numbers: is it real (region-level FDR), and how robust is it to the analyst's parameter choices.

WHAT IS DIFFERENT ABOUT IT

Filtering that costs nothing Keeping only LD-supported markers before correcting for multiple testing is <i>independent filtering</i> : it raises power by shrinking the tested set without breaking error control, because local LD is independent of the association statistic under the null.	Regions, not SNPs Markers are grouped by linkage rather than by physical distance, and the joining threshold self-calibrates to each chromosome's LD decay. Multiple testing is then reported over tens of regions, not hundreds of thousands of SNPs.	No cherry-picking Every analyst-chosen parameter is integrated over instead of fixed, and the reported score says how much of that parameter space supports the locus. A result that holds in only one corner of the grid is visible as such.
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THE IDEA IN ONE PICTURE



HOW IT DIFFERS FROM EXISTING APPROACHES

Haplotype and LD-pattern sweep scans [H12](#) ·

[OmegaPlus](#)

These read elevated LD as direct evidence of selection. LDscnR does not — elevated LD also arises from inversions, recombination-rate variation, demography and structure. Local LD is used only as an *independent* measure of genomic coherence for prioritising association signals; it is never interpreted as a test of selection and never folded into the evidence.

LD-based complexity reduction [the LDna lineage](#)

The consistency score originates here, not with LDscnR. Fang *et al.* (2021) ran **144 tests** on this same data — three LDna clustering parameters × four multiplicity and inflation corrections, one of them the permutation of Li *et al.* (2018), affordable only because complexity reduction had shrunk the test count — and scored each region by the proportion of tests finding it significant, discarding $C < 0.05$. LDscnR keeps that principle and changes what it applies to: LD *filters* which SNPs are tested rather than replacing them with cluster-level units, the sweep runs over continuous nuisance parameters at two nested scales, and the structure-aware null is promoted from one correction among four to the inferential instrument. It is also **the first time the consistency principle is benchmarked against simulations with known causal variants**.

Window-based regional aggregation [WZA](#)

Fixed physical windows assume adjacent variants carry one evolutionary signal; recombination means they often do not, so aggregation dilutes highly localised signal. LDscnR takes regions from the topology of the local LD network — overlapping but distinct signals in the same interval stay distinct — and preserves the original SNP-level statistics instead of summarising them.

How significance is established

Permutation is the acknowledged gold standard for multiple-testing correction, because it accounts for the correlation structure of the genome directly. Under population structure the standard version is invalid — reshuffling phenotypes destroys the among-individual correlation the model assumes — and the correct substitute is already specified: draw surrogate phenotypes from $MVN(0, \sigma_g^2 A + \sigma_e^2 I)$ and rerun the scan on each. What limits it is cost, and cost scales with the number of tests. The lineage bought affordability by *complexity reduction* — testing cluster principal components or multi-locus alleles instead of SNPs — which shrinks the test count but tests a unit weaker than the strongest SNP in the region. **LDscnR instead filters, keeps the per-SNP statistics, and moves inference to the region**, so the null need only resolve a handful of regions per surrogate: 50–200 surrogates rather than tens of thousands.

STATUS

Validated on forward-in-time, individual-based simulations (NEMO) with known causal variants, and applied to the three-spined stickleback marine-freshwater panel of Fang *et al.* (2021) — 117 individuals, 35 populations, 47–75°N — where single-SNP scans resolve nothing. There it recovers the known architecture — the chromosome-1 inversion, the *Eda* region, and peaks on chromosomes 5, 7, 17 and 20 — with the regions independently corroborated against a published set of recurrent marine-freshwater adaptation peaks from separate cohorts and a different assembly.

Implemented as an R package: github.com/genomeinsights/LDscnR. Manuscript in preparation — comments welcome at petri@genomeinsights.fi.